Passage 2: Databases: Some Real-Life Examples

Databases are very important in modern computer systems. They help us store, organize, and find data easily. A Database Management System (DBMS) is software that allows users and programs to access databases. There are many types of databases. Let’s look at how different types of databases store data, and how different DBMS manage the databases.

TYPES OF DATABASES

Disk-Based Databases

Disk-based databases are the classic kind of database. They store data on a hard drive or SSD. In this way the data can be kept safely for a long time. Many companies use this type because it is reliable. For example, Meta uses disk-based databases to store user data for Facebook and Instagram. The data is kept safe even after a system shutdown. Developers choose this type when they need safe and permanent storage.

In-Memory Databases

Another type is the in-memory database. This database stores data in RAM, which is much faster than disk storage. Because of this, it can process data very quickly. However, the data is usually temporary. Twitter uses *Redis* to store and quickly load user timelines. Developers use this type when speed is very important.

Embedded Databases

Embedded databases are built inside an application, so they do not need a separate server. They are often used in mobile apps. For example, the messaging app, Whatsapp, uses SQLite to store messages on a user’s phone, which can keep the messages more private. This also allows the user to access data without the internet. Developers choose this type for simple, private systems and offline use.

Columnar Databases

Some databases are made for special tasks. Columnar databases store data in columns instead of rows. This makes them useful for analyzing large amounts of data. Amazon, for example, uses Redshift to study business data and create reports. These databases are useful when companies want to understand trends and patterns.

Graph Databases

Graph databases are designed to show relationships between data. They use data points, called ‘nodes’, and connections between the nodes, called ‘edges’, much like a network. This is helpful for systems like social networks. LinkedIn uses graph technology to show how users are connected. Amazon also uses similar a graph database to recommend products. Developers choose graph databases when relationships between different data sets are very important.

Time-Series Databases

Time-series databases store data with time information. This means each piece of data has a timestamp. These databases are useful for systems that collect data over time. For example, PayPal uses them to monitor system performance and transactions. Developers use this type when they need to track changes over time.

TYPES OF DATABASE MANAGEMENT SYSTEMS

Whatever type of database you use, you also need a Database Management System, or DBMS. There are many popular DBMS, like MySQL, Oracle, and Microsoft SQL. Most DBMS are either ‘relational’, or ‘non-relational’.

Relational DBMS

The relational DBMS is the most common type. It stores data in tables with rows and columns. This type uses Structured Query Language (SQL) and has a fixed structure of relationships between entities (pieces of data) and variables. It is very reliable and accurate. Relational databases are used in many situations, like banking and shopping systems.

Non-Relational DBMS

Other DBMS are non-relational, also called NoSQL. The advantage of NoSQL is that it is more flexible. It is easy to make changes, and is better for scalability (scaling the database to a larger size). This is because NoSQL does not need a fixed structure, and can manage large amounts of changing data in different formats, such as JSON or key-value pairs. For example, Instagram uses the NoSQL called Cassandra to manage large amounts of data, and Amazon uses one called DynamoDB for scalable systems.

Relational and non-relational DBMS are similar in many ways. They both store and manage data, and both allow users to quickly search and update data. The main difference is that relational databases are more structured, while NoSQL databases are more flexible. In general, relational systems are better for accuracy, and non-relational systems are better for **scalability**.

In many real-world systems, companies use more than one type of database. For example, they may use a relational database for important data, a NoSQL database for large data, and an in-memory database for quick, temporary tasks. This helps companies build systems that are both fast and reliable.

A useful database depends on these two factors: type of storage and how it is managed. The storage type you want to use depends on what task you want the database to perform, while the types of DBMS give you different ways to organize the data. By understanding the types of storage and DBMS, you can choose the best database setup for your needs.

SOURCES:

DEV Community - Real-World DBMS Examples: Explained with Use Cases (2026)
https://dev.to/rishabhtpt/real-world-dbms-examples-explained-with-use-cases-2omj

Medium.com - Curious About Databases? Here’s How Big Tech Uses Them (with Real Examples)
https://medium.com/@sauravx25/curious-about-databases-heres-how-big-tech-uses-them-with-real-examples-876b772d8e02